

SOC Analyst's Notebook

Windows Event Log Analysis — Field Reference

By Satya Upendra Samana | SOC L1 Learning Journey | July 2026

This is where many beginners go wrong. They think documentation means theory — a textbook that explains what an operating system is, what a service does, what Active Directory means. That is foundation knowledge, not analyst knowledge.

This notebook is different. It is written the way a real SOC analyst would write their investigation notes — what you observed, what you did, why you did it, and what it means. Every topic here maps to a real decision a junior analyst must make when staring at a Windows Security log at 2 AM.

If you are new to this, read it like a manual for the job, not a textbook for an exam.

1. The Security Log — Your Primary Source

Windows keeps three main logs: System (OS health), Application (individual software), and Security (everything access-related). As a SOC analyst, you live in the Security log. Almost every investigation starts here.

To open it: Event Viewer → Windows Logs → Security. To filter by Event ID: right-click Security → Filter Current Log → type the Event ID number.

To pull it via PowerShell — faster when you know exactly what you are looking for:

```
Get-WinEvent -LogName Security | Where-Object {$_.Id -eq 4625}
```

One thing that catches every beginner: 'Audit Success' does not mean something safe happened. It means Windows successfully recorded the event. A hacker logging in with a stolen password also shows Audit Success — because the login worked. Success versus safe are two different things.

2. Anatomy of a Log Entry — What Every Field Means

Every log entry you will ever read has the same structure. Before you can investigate anything, you need to read a single entry fluently.

Field	What it tells you
Date and Time	When the event happened — the backbone of every timeline
Event ID	A numbered code for the type of action (e.g. 4624 = successful logon)
Keywords	Audit Success or Audit Failure — did the audited action complete or fail
Account Name	Who the event is about (the target account)
Subject	Who performed the action (the actor — could be a user, a service, or SYSTEM)

Field	What it tells you
Logon Type	How the logon happened — physical keyboard, network, RDP, service startup
Source Network Address	Where the request came from — 127.0.0.1 means local, anything else is a remote address
Logon ID	A unique session identifier — use this to match a logon (4624) to its logoff (4634/4647)

3. The Core Event IDs — What Every SOC L1 Analyst Must Know

You do not need to memorise hundreds of Event IDs. These seven cover the majority of real SOC L1 investigations. Know them cold.

Event ID 4624 — Successful Logon

Someone (or something) successfully logged into this machine. This fires for human logins, background service logins, and everything in between. The Logon Type field tells you which kind.

Analyst eye: Never read 4624 alone. Check the Logon Type (was it remote?), the Source Network Address (where did it come from?), and the time (is this normal working hours?). A 4624 after several 4625 failures is the most important combination in SOC work.

Event ID 4625 — Failed Logon

A logon attempt failed — wrong password, account locked, account doesn't exist. This is the most watched Event ID in SOC environments because repeated failures are the signature of a brute-force attack.

Analyst eye: One or two 4625 entries for a real user account is probably a typo. Five or more in quick succession against the same account is a pattern. Forty in two minutes followed by a 4624 is a confirmed brute-force breach. Count the entries, check the timing, look at what comes after.

Event ID 4634 / 4647 — Logoff

A login session ended. 4634 is generated by the system when it closes a session. 4647 is generated when the user themselves clicked Sign Out. Both mean the same thing: the session is over.

Analyst eye: Match the Logon ID in a 4634/4647 entry to the Logon ID in its corresponding 4624 entry. That gives you the exact session duration. A session that lasted two minutes at 3 AM from an unusual account is very different from a normal 8-hour workday session.

Event ID 4720 — New User Account Created

A new local user account was created on this machine. This is normal when IT onboards a new employee — but it is deeply suspicious outside of that context, especially if no one requested it.

Analyst eye: The first question is always: who created it (check the Subject field) and is that expected? The second question is what the account name looks like — attackers deliberately choose names that blend into the system, like svc_backup or WinHelper, not hacker123. A new account created at 2 AM with no IT ticket is a red flag.

Event ID 4732 — User Added to Security Group

An existing account was added to a local security group. The group that matters most is Administrators — the built-in group that grants full control of the machine. This is the event that marks privilege escalation.

Analyst eye: The moment backdoor_svc or any unexpected account appears in the Administrators group, the attacker has full control. This is the single highest-value event to alert on in a local machine investigation. Catch this before the attacker installs their persistence mechanism and you have stopped most of the damage.

Event ID 4688 — New Process Created

A new program started running. This event includes not just what ran, but what started it — the parent process. That parent-child relationship is where the real intelligence sits.

Analyst eye: Normal: explorer.exe launching notepad.exe. Suspicious: winword.exe launching powershell.exe (a Word document should never need to open a scripting tool). Also watch for processes running from unusual locations — a legitimate Windows tool running from C:\Users\Public\ or a Temp folder is a red flag regardless of the filename.

Event ID 4697 — Service Installed

A new Windows service was registered on this machine. Services run in the background with high privilege (often as LocalSystem, the most powerful account on Windows) and survive reboots — which is exactly why attackers use them for persistence.

Analyst eye: Two things to check immediately: the service name (does it look like it is impersonating a real Windows component?) and the file path (legitimate services almost never run from user-writable folders like C:\Users\Public\). This event requires a specific audit subcategory to be enabled — Security System Extension must be set to Success and Failure in auditpol, otherwise 4697 is silently dropped even when a service is created.

4. Logon Types — How Someone Got In Matters as Much as Whether They Got In

The Logon Type field sits inside every 4624 and 4625 event. It is one of the highest-value fields a SOC analyst reads because the same successful login can be completely normal or deeply suspicious depending on how it happened.

Type	Name	What it means	Risk level
2	Interactive	Physically at the keyboard — someone typed a password at this machine	Normal
3	Network	Accessing a shared folder or printer over the local network	Usually normal — check source IP
5	Service	A Windows service logged in automatically (SYSTEM, LocalSystem, etc.)	Normal — expected system noise
7	Unlock	An already-logged-in session was unlocked	Normal

Type	Name	What it means	Risk level
8	NetworkCleartext	Network login with password sent unencrypted	⚠ Suspicious — modern systems should not do this
10	RemoteInteractive	Remote Desktop (RDP) — someone is controlling this machine from another computer	⚠ High attention — most commonly abused by attackers
11	CachedInteractive	Logged in with cached credentials when the Domain Controller was unreachable	Situational

The combination that matters most: Logon Type 10 (RDP) + an external public IP address + odd hours + an account that never normally uses RDP = immediate escalation to L2. This combination is how the majority of real ransomware attacks start.

5. Correlation — How You Build the Story

A single log entry rarely tells you what happened. SOC analysis is connecting multiple entries together, ordered by time, to reconstruct the full sequence of events. This is called correlation.

The four fields you correlate on:

- Logon ID — connects a logon (4624) to its matching logoff (4634/4647) to calculate session duration
- Account Name / SID — connects everything one specific user did across different Event IDs
- Source Network Address — connects multiple events from the same origin, even across different accounts
- Time window — groups events that happened close together to identify related activity

The attack chain from this investigation, as a correlation example:

```
4625 x9 → 4624 (same account, 21 seconds later) → 4720 (new account created) → 4732 (new account added to Administrators) → 4697 (service installed)
```

Each entry alone looks like a single event. Connected in order, they tell the complete story of a brute-force attack followed by privilege escalation and persistence.

6. Attack Patterns — What the Logs Look Like When Something Is Wrong

Brute Force

What it looks like: many 4625 entries in rapid succession against the same account, often followed by a 4624. The speed matters — a real human mistyping takes seconds between attempts; an automated tool fires dozens per minute.

The key question: did a 4624 follow? If yes, the attacker got in. That changes the priority from detection to containment.

Privilege Escalation

What it looks like: a 4732 event showing a standard user account or a newly created account added to the Administrators group. The subject field shows who did it. If it was not a known IT admin account performing a planned change, treat it as hostile.

Persistence

What it looks like: 4720 (new account) followed by 4732 (added to Administrators) followed by 4697 (new service) — all within a short time window. Each piece alone might be explainable. All three together in a short window is the persistence playbook.

Lateral Movement

What it looks like: the same account appearing in 4624 entries across multiple different computers in a short time span, using Logon Type 3 (network) or Type 10 (RDP). A real employee logs into one or two machines. An attacker who has stolen domain credentials logs into many.

7. The Gap Nobody Talks About — Audit Policy

Here is something that will catch you in a real investigation: Windows does not log every security-relevant event by default. Specific subcategories of auditing are disabled out of the box, which means evidence can silently disappear even when the event genuinely happened on the machine.

The two gaps found during this investigation:

- Security System Extension was set to No Auditing — Event ID 4697 (service installed) was completely suppressed. The service was created successfully but left no log entry until the subcategory was manually enabled.
- Account lockout threshold was 10 attempts, not 5 — a simulation designed around a 5-attempt lockout never triggered a 4740 (account lockout) event at all.

Before you conclude that something did not happen because there is no log for it, check the audit policy:

```
auditpol /get /subcategory:"Security System Extension"  
auditpol /set /subcategory:"Security System Extension" /success:enable /failure:enable
```

In a real SOC environment, audit policy is configured centrally via Group Policy on the Domain Controller and pushed to every machine in the domain. On a standalone Workgroup machine like a personal laptop, it has to be set manually. Either way, verifying it before you start investigating is not optional.

8. How to Use This Notebook

This is not a document you read once. It is a reference you return to when you are in the middle of an investigation and need to remember what a field means, why a Logon Type matters, or which Event ID to filter for next.

The fastest way to improve as a SOC L1 analyst is not to memorise more theory. It is to investigate more logs. Generate real evidence on a machine you control, filter for the Event IDs above, build a timeline from what you find, and write down what the story tells you. Do that enough times and pattern recognition becomes instinct.

That is what this project is — and that is what every entry in this notebook came from.

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